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Step B to C: Quasi-static adiabatic expansion from (P_2,V_2,T_H) to (P_3,V_3,T_L)

The cylinder is placed on the insulating stand and the piston is allowed to move out. As the gas expands adiabatically from volume V_2 to volume V_3 the pressure falls from P_2 to P_3. The temperature falls to T_L. This adiabatic expansion is represented by curve *BC* in the P-V diagram. This adiabatic process also occurs quasi-statically and implying that this process is reversible and the ideal gas is in equilibrium throughout the process. It is shown in the figure 8.47(b). From the equation (8.42)

The work done by the gas in an adiabatic expansion is given by,

$$W_{B\to C} = \int_{V_2}^{V_3} PdV = \frac{\mu R}{\gamma - 1}\left[T_H - T_L\right] =$$

Area under the curve BC (8.57)

This is shown in Figure 8.46 (b)

Step C → D: Quasi-static isothermal compression from (P_3,V_3,T_L) to (P_4,V_4,T_L): It is shown in the figure 8.47(c)

The cylinder is placed on the sink and the gas is isothermally compressed until the pressure and volume become P_4 and V_4 respectively. This is represented by the curve *CD* in the PV diagram as shown in Figure 8.45. Let $W_{C\to D}$ be the work done on the gas. According to first law of thermodynamics

$$\therefore W_{C\to D} = \int_{V_3}^{V_4} PdV = \mu RT_L \ln\left(V_4/V_3\right) = -\mu RT_L \ln\left(\frac{V_3}{V_4}\right)$$

. = - Area under the curve CD (8.58)

This is shown in Figure 8.46 (c)

Here V_3 is greater than V_4. So the work done is negative, implying work is done on the gas.

Step D→A: Quasi-static adiabatic compression from (P_4,V_4,T_L) to (P_1,V_1,T_H): It is shown in the figure 8.47(d)

The cylinder is placed on the insulating stand again and the gas is compressed adiabatically till it attains the initial pressure P_1, volume V_1 and temperature T_H. This is shown by the curve *DA* in the P-V diagram.

$$\therefore W_{D\to A} = \int_{V_4}^{V_1} PdV = \frac{\mu R}{\gamma - 1}\left(T_L - T_H\right) = -\text{ Area}$$

under the curve DA (8.59)

In the adiabatic compression also work is done on the gas so it is negative, as is shown in Figure 8.46 (d)

Let 'W' be the net work done by the working substance in one cycle

∴ W=Work done by the gas – work done on the gas

$$= W_{A\to B} + W_{B\to C} - W_{C\to D} - W_{D\to A}$$

since $W_{B\to C} = W_{D\to A}$

$$= W_{A\to B} - W_{C\to D}$$

The net work done by the Carnot engine in one cycle $W = W_{A\to B} - W_{C\to D}$ (8.60)

Equation (8.60) shows that the net work done by the working substance in one cycle is equal to the area (enclosed by ABCD) of the P-V diagram (Figure 8.48)

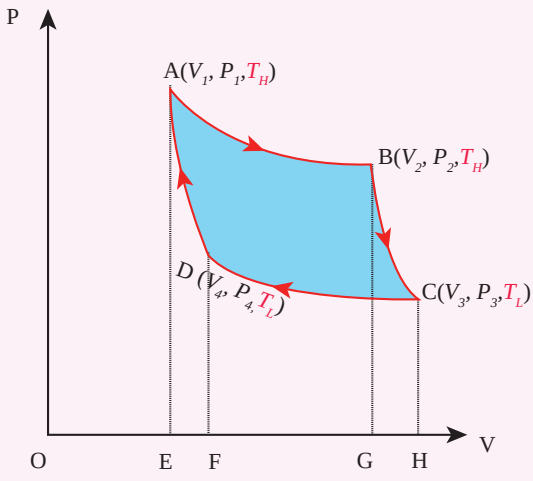


Figure 8.48 Net work done in Carnot cycle